Linear Discrimination

$$g(x) = w^T \cdot x + w_0$$

by assuming $p(x|y=1) = N(x; \mu_1, \Sigma)$, $p(x|y=2) = N(x; \mu_2, \Sigma)$

$\hat{\mu}_1$ → sample mean of class1

$$\underset{D\times 1}{\hat{w}} = \hat{\Sigma}^{-1} \cdot (\hat{\mu}_1 - \hat{\mu}_2)$$

$\hat{\mu}_2$ → sample mean of class2

$$\underset{1\times 1}{\hat{w}_0} = -\frac{1}{2}(\hat{\mu}_1 + \hat{\mu}_2)\hat{\Sigma}^{-1}(\hat{\mu}_1 - \hat{\mu}_2) + \log\left[\frac{\hat{Pr}(y=1)}{\hat{Pr}(y=2)}\right]$$

$\hat{\Sigma}$ → sample covariance of all data points

$\hat{Pr}(y=1)$ → frequency of class 1

$\hat{Pr}(y=2)$ → frequency of class 2

$$\exp\left[\log\left[\frac{\delta}{1-\delta}\right]\right] = \exp[w^T \cdot x + w_0]$$

$$\frac{\delta}{1-\delta} = \exp[w^T \cdot x + w_0] \Rightarrow \delta = \exp[w^T \cdot x + w_0] - \delta\exp[w^T \cdot x + w_0]$$

$$\delta + \delta\exp[w^T \cdot x + w_0] = \exp[w^T \cdot x + w_0]$$

$$\delta(1 + \exp[w^T \cdot x + w_0]) = \exp[w^T \cdot x + w_0]$$

$$\delta = \frac{\exp[w^T \cdot x + w_0]\exp[-(w^T \cdot x + w_0)]}{1 + \exp[w^T \cdot x + w_0]\exp[-(w^T \cdot x + w_0)]}$$

$$\delta = \frac{1}{1 + \exp[-(w^T \cdot x + w_0)]}$$

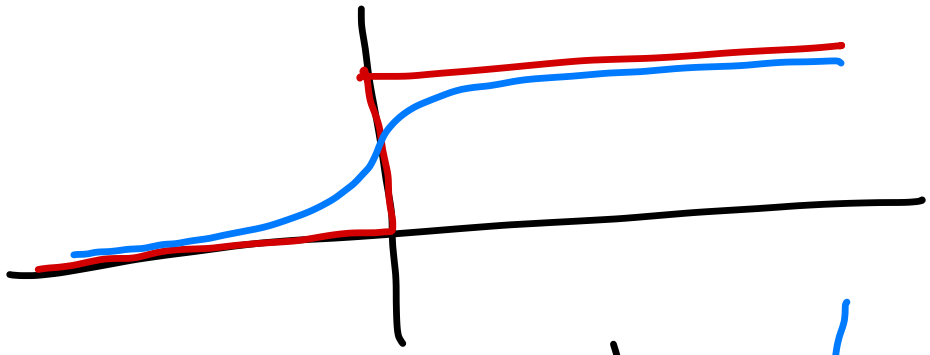
Sigmoid

i) $w^T.x + w_0 > 0 \Rightarrow \delta > 0.5$ $\quad 1-\delta < 0.5$

ii) $w^T.x + w_0 = 0 \Rightarrow \delta = 0.5$ $\quad 1-\delta = 0.5$

iii) $w^T.x + w_0 < 0 \Rightarrow \delta < 0.5$ $\quad 1-\delta > 0.5$

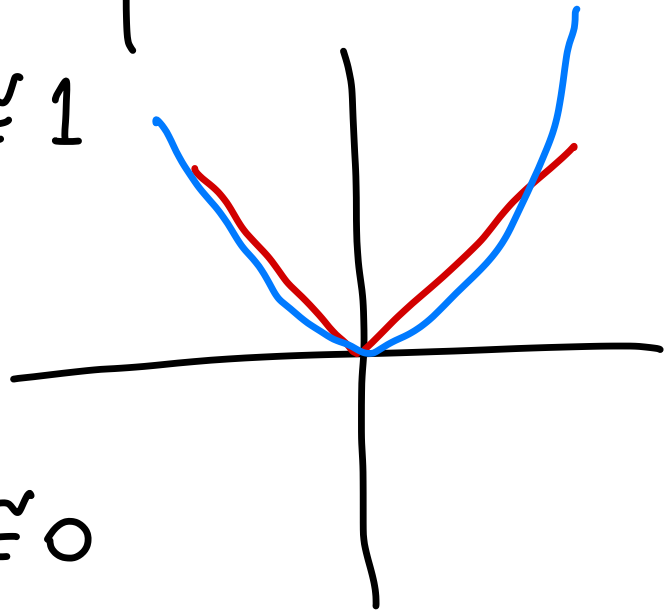
$$\delta(a) = \frac{1}{1+\exp[-a]}$$



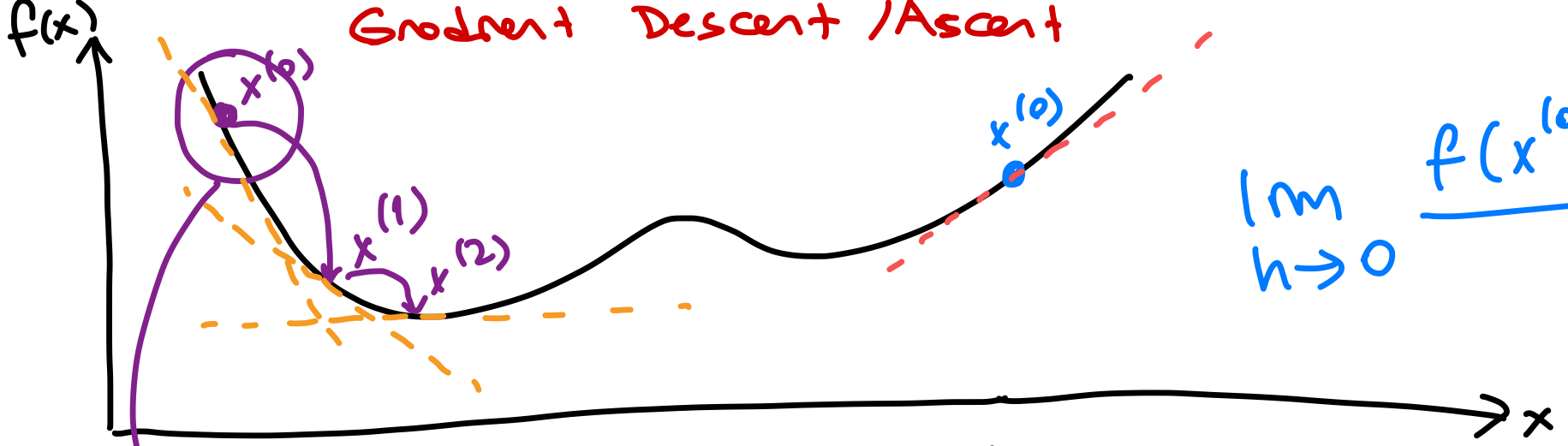
if $a=+5 \Rightarrow \frac{1}{1+\exp(-5)} \simeq 0.9933 \simeq 1$

if $a=0 \Rightarrow \frac{1}{1+\exp(0)} = 0.5$

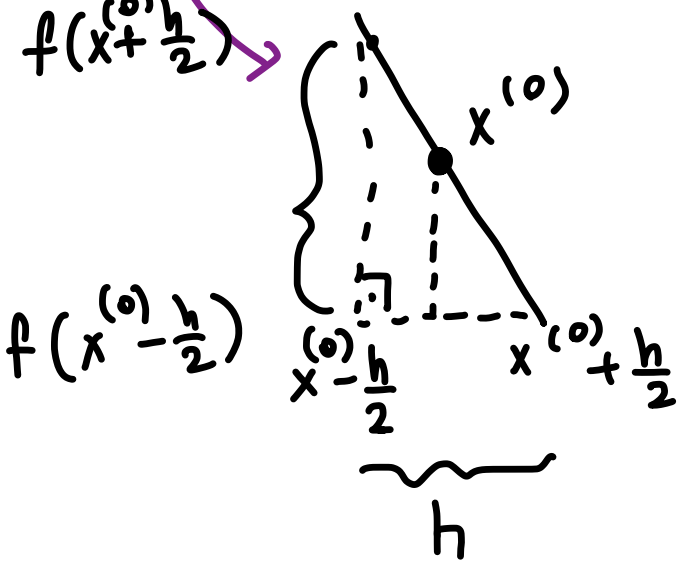
if $a=-5 \Rightarrow \frac{1}{1+\exp(+5)} \simeq 0.0067 \simeq 0$



Gradient Descent / Ascent



$$\lim_{h \to 0} \frac{f(x^{(0)}+h) - f(x^{(0)})}{h}$$



$$\lim_{h \to 0} \frac{f(x^{(0)}+\frac{h}{2}) - f(x^{(0)}-\frac{h}{2})}{h}$$

stepsize

$$\Delta x = -\eta \cdot \frac{\partial f(x)}{\partial x}$$

$$x^{(t+1)} = x^{(t)} - \eta \cdot \frac{\partial f(x)}{\partial x}$$

update equation

$$(w^*, w_0^*) = \underset{(w,w_0)}{\arg\min}\ E[w, w_0 \mid \mathcal{X}]$$

↳ training set

$$\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N} \qquad x_i \in \mathbb{R}^D \qquad y_i \in \{0, 1\}$$

failure ← 0, 1 → success

$$y_i | x_i \sim \text{Bernoulli}\left(y_i;\ \hat{Pr}(y_i = 1 \mid x_i)\right)$$

$\hat{y}_i$ → predicted posterior probability.

$$\text{Likelihood}(w, w_0 \mid \mathcal{X}) = \prod_{i=1}^{N} \left[\hat{y}_i^{\,y_i} \cdot (1 - \hat{y}_i)^{1 - y_i}\right]$$

$$\log \text{likelihood}(w, w_0 \mid \mathcal{X}) = \sum_{i=1}^{N} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)\right]$$

$$E[w, w_0 \mid \mathcal{X}) = -\sum_{i=1}^{N} \left[\underbrace{y_i}_{\text{constant}} \underbrace{\log(\hat{y}_i)}_{f(w,w_0)} + \underbrace{(1 - y_i)}_{\text{constant}} \underbrace{\log(1 - \hat{y}_i)}_{f(w,w_0)}\right]$$

negative log likelihood / error / loss

minimize $-\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right]$

with respect to w, w_0

$$\hat{y}_i = \frac{1}{1+\exp\left[-(w^T x_i + w_0)\right]}$$

$$\frac{\partial E[w, w_0 | \mathcal{X}]}{\partial w} = ? \qquad \frac{\partial E[w, w_0 | \mathcal{X}]}{\partial w_0} = ?$$

$$\frac{\partial \log(\hat{y}_i)}{\partial w} = \frac{\partial \log(\hat{y}_i)}{\partial \hat{y}_i} \frac{\partial \hat{y}_i}{\partial w}$$

$$= \frac{1}{\hat{y}_i} \cdot \hat{y}_i (1-\hat{y}_i) \cdot x_i$$

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$$= (1-\hat{y}_i) \cdot x_i$$

$$\frac{\partial \log(a)}{\partial a} = \frac{1}{a}$$

$$\frac{\partial \frac{f(a)}{g(a)}}{\partial a} = \frac{\frac{\partial f(a)}{\partial a} g(a) - \frac{\partial g(a)}{\partial a} f(a)}{[g(a)]^2}$$

$$\text{sigmoid}(a) = \frac{1}{1+\exp(-a)}$$

Show that $\frac{\partial\, \text{sigmoid}(a)}{\partial (a)} = \text{sigmoid}(a)\left[1 - \text{sigmoid}(a)\right]$

$$\frac{\partial\, \text{sigmoid}(a)}{\partial\, a} = \frac{\frac{\partial\, 1}{\partial\, a}\cdot\left[1+\exp(-a)\right] - \frac{\partial\left(1+\exp(-a)\right)}{\partial\, a}\cdot 1}{\left[1+\exp(-a)\right]^2}$$

($\frac{\partial\, 1}{\partial\, a} = 0$)

$w^T \cdot x = x^T \cdot w$

$$= \frac{1\,(-1)\exp(-a)}{\left[1+\exp(-a)\right]^2} = \underbrace{\frac{1}{1+\exp(-a)}}_{\text{sigmoid}(a)} \cdot \underbrace{\frac{\exp(-a)}{1+\exp(-a)}}_{\left[1-\text{sigmoid}(a)\right]}$$

$$\frac{\partial\, \text{sigmoid}(\overbrace{w^T \cdot x_i + w_0}^{a})}{\partial\, w} = \frac{\partial\, \text{sigmoid}(a)}{\partial\, a}\,\frac{\partial\, a}{\partial\, w}$$

$$= \underbrace{\text{sigmoid}(w^T \cdot x_i + w_0)}_{\hat{y}_i}\cdot\left[\underbrace{1 - \text{sigmoid}(w^T \cdot x_i + w_0)}_{(1-\hat{y}_i)}\right]\cdot \underbrace{x_i}_{x_i}$$

Exercise $\frac{\partial \log(\hat{y}_i)}{\partial w_0} = (1-\hat{y}_i)$

Exercise $\frac{\partial \log(1-\hat{y}_i)}{\partial w} = ?$ $\frac{\partial \log(1-\hat{y}_i)}{\partial w_0} = ?$

$$\frac{\partial Error[w, w_0 | \mathcal{X}]}{\partial w} = -\sum_{i=1}^{N} \left[y_i (1-\hat{y}_i) \cdot x_i + (1-y_i) \boxed{\quad} \right]$$

$$= \boxed{-\sum_{i=1}^{N} (y_i - \hat{y}_i) \cdot x_i}$$

$$\frac{\partial Error[w, w_0 | \mathcal{X}]}{\partial w_0} = \boxed{-\sum_{i=1}^{N} (y_i - \hat{y}_i)}$$

$$w^{(t+1)} = w^{(t)} - \eta \cdot \frac{\partial Error}{\partial w}$$

$$w_0^{(t+1)} = w_0^{(t)} - \eta \cdot \frac{\partial Error}{\partial w_0}$$

